Fiscal Year:	FY 2005	**Task Last Updated:**	FY 08/16/2005
PI Name:	Hadjiargyrou, Michael Ph.D.		
Project Title:	Elucidating the Molecular Events of Bone Mechanoreception		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	None		
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City:	Stony Brook	**State:**	NY
Zip Code:	11794	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
Start Date:	08/15/2001	**End Date:**	08/14/2005
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rubin, Clinton (State University of New York) Dhundale, Anil (Stony Brook)		
Grant/Contract No.:	NAG 2-1517		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	It is clear that changes in gravitational forces (i.e., microgravity, hypergravity) will readily affect the form and function of living organisms. However, the molecular mechanisms by which physical stimuli influence the cells, tissues, organs and organisms remain unknown. This goal is particularly pertinent in the skeletal system, where the osteopenia that results from microgravity may hamper an extended human presence in space. With the recent technological advances in molecular biology, it is now possible to investigate the complex regulatory mechanisms responsible for the physical control of bone mass and morphology. We believe that robust changes (activation/inactivation) of mechanically sensitive gene transcription is ultimately responsible for regulating both the resorptive response to decreases in mechanical stimuli, and the anabolic response to increases in mechanical stimuli. We will use our unique cDNA library of bone expressed genes, our unique anabolic mechanical stimulus, and two specific strains of mice, one of which is responsive to changes in mechanical stimuli and one of which is not, to address these complex questions. In the first phase of this proposal, we will investigate the differential gene response of these two distinct strains of mice to disuse, to identify early transcriptional activity that leads to bone resorption. In the second phase, we will use the same animal model to isolate and identify early response genes following exposure to a strong osteogenic stimulus. The third phase of the study will identify transcriptional regulators common to both formation and resorption of bone, with the goal of, in the fourth phase, over-expressing one of these regulators in mice that do not respond to changes in mechanical stimuli to determine if the bone tissue becomes responsive. These studies will provide unique insight into the molecular mechanisms responsible for bone's sensitivity to mechanical forces, and may help define etiologic factors that contribute to the incidence of bone resorption and osteopenia. To provide unique insight into the molecular mechanisms responsible for bone's sensitivity to mechanical forces. May help define etiologic factors that contribute to the incidence of bone resorption and osteopenia.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We strongly believe that in the context of human health, the proposed experiments will help to determine mechanistic causes of generalized earth or space-based osteopenia and reveal the potential therapeutic use of gene products themselves or as appropriate drug targets for beneficial clinical intervention.
Task Progress:	As a result of this completed grant, our data indicate that the skeletons of distinct inbred strains respond differentially to disuse at the tissue level, a disparate morphologic response which correlates strongly with early expression patterns. Given the relatively small number of candidate genes screened in this study, but the high percentage of them that were influenced by the lack of weight bearing, the total number of genes involved in these processes appears to be high, yet the transcriptional changes are subtle and interdependent. When considering bone’s response to changes in its functional environment in vivo, it appears that the adaptive process is much more subtle than might be expected in tissue repair. Clearly, the strong effects of genetic variations will have to be considered for advancing diagnostics and therapeutics of osteoporosis, yet the strong correlation in altered gene expression patterns across genomes indicates that pharmaceutical interventions may be able to take advantage of similar molecular pathways.
Bibliography Type:	Description: (Last Updated: 09/10/2012)
Articles in Peer-reviewed Journals	Judex S, Zhong N, Squire ME, Ye K, Donahue LR, Hadjiargyrou M, Rubin CT. "Mechanical modulation of mechanical signals which regulate anabolic and catabolic activity in bone tissue." J Cell Biochem. 2005 Apr 1;94(5):982-94. PMID: 15597385 , Apr-2005
Articles in Peer-reviewed Journals	Zhong N, Garman RA, Squire ME, Donahue LR, Rubin CT, Hadjiargyrou M, Judex S. "Gene expression patterns in bone after 4 days of hind-limb unloading in two inbred strains of mice." Aviat Space Environ Med. 2005 Jun;76(6):530-5. PMID: 15945395 , Jun-2005